TLP:AMBER

HTOC

HEALTHCARE THREAT OPERATIONS CENTER

ThreatConnect Playbook — Get CDN Scanning Indicators by ASN

Version 1.0
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1. Purpose

This ThreatConnect playbook automatically identifies and labels **CDN-hosted scanning indicators** within HTOC's own indicator set. It queries for Address indicators belonging to the Linode/Akamai ASN (63949) that have not yet been labeled, applies the `CDN PB` (**CDN Playbook**) tag to mark them as CDN-sourced scanning activity, and stores the indicator data in the ThreatConnect DataStore. Email alerts are sent on any failures.

The `CDN PB` tag is the primary identifier that an indicator has been reviewed and classified by this playbook as originating from CDN infrastructure. The TQL exclusion filter (`NOT hasTag("CDN PB")`) ensures each indicator is only processed once. The playbook runs twice daily to keep labeling current as new indicators are added to ThreatConnect.

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2. Scope

This procedure applies to HTOC analysts and ThreatConnect administrators who monitor, trigger, modify, or troubleshoot this playbook. It covers the playbook's trigger configuration, step-by-step logic, error handling, and maintenance guidance.

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3. Prerequisites

Requirement	Notes
ThreatConnect access	Must have playbook view/execute permissions
HTOC Org indicators	Address indicators with ASN 63949 must be present under the `HTOC Org` owner in ThreatConnect
ThreatConnect DataStore	Local DataStore must be available for POST operations
Email server configured	SMTPS (Implicit TLS) connection required for failure alert emails
Notification recipient	`jaytlin.askew@hhs.gov` — update if ownership changes

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4. Trigger Configuration

The playbook has three defined triggers. The **Every 12 Hours Trigger** is the current active production trigger. The WebHook Trigger is used for testing and manual execution only.

Trigger Name	Type	Schedule	Status
Every 12 Hours Trigger (6AM-6PM)	Timer	6:00 AM and 6:00 PM (`0 0 6,18 * * ?`)	**Active — Production**
WebHook Trigger	HttpLink (Webhook)	On demand	Testing / manual use only

Trigger Name	Type	Schedule	Status
Every 1 Minute Trigger (Testing Trigger)	Timer	Every 1 minute (`0 * * * * ?`)	Inactive

Every 12 Hours Trigger (Production)

Type: Timer

Schedule: Cron `0 0 6,18 * * ?` — fires at **6:00 AM and 6:00 PM** daily

Purpose: Ensures new CDN scanning indicators are tagged and stored twice daily without manual intervention

WebHook Trigger (Testing / Manual)

Type: HTTP Link (external webhook)

Timeout: 5 minutes

Cache: 120 minutes (prevents duplicate executions within the cache window)

HTTP Response Body: Returns the success log message (`logger.content`) from the final success step

Purpose: Used for testing changes to the playbook or triggering a manual run outside the normal schedule. Send an HTTP request to the webhook URL found in the ThreatConnect playbook editor to invoke it.

> **Note:** Do not leave the WebHook Trigger as the primary active trigger in production. The 12-hour timer is the correct production trigger.

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5. TQL Query

The playbook queries ThreatConnect using the following TQL expression in the **ThreatConnect Get Indicators** step:

```
(ownerName in ("HTOC Org"))
AND typeName in ("Address")
AND addressASN IN (63949)
AND NOT hasTag(summary IN ("CDN PB"))
```

Filter	Value	Purpose
Owner	\`HTOC Org\`	Limits to HTOC's own indicator set
Type	\`Address\`	IP addresses only
ASN	63949	Targets Linode / Akamai CDN hosting range
Tag exclusion	\`NOT hasTag("CDN PB")\`	Skips indicators already processed by this playbook
Max results	1	Processes one indicator per execution

> **Important:** The `NOT hasTag("CDN PB")` filter is what prevents reprocessing. If this tag is removed from an indicator, the playbook will process it again on the next scheduled run.

Monitored ASN

ASN	Known Provider
63949	Linode / Akamai

To add additional ASNs, update the `addressASN IN (...)` clause in the TQL field of the **ThreatConnect Get Indicators** step and add a comma-separated list of ASN numbers.

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6. Playbook Flow

6.1 Error Paths

Failure Point	Behavior
Get Indicators fails	→ Logger: "Get Indicators FAILED. Please check the TQL and summary for output result errors." → Email: subject "GET Indicator Process Failed", body = logger content, to \`jaytlin.askew@hhs.gov\`
Iterator fails (no results returned)	→ Email: subject "Get CDN Scanning Indicators by ASN Failed.", body = static message, to \`jaytlin.askew@hhs.gov\`
Set HTOC Scanner Identifier (tag add) fails	→ Logger: "Process Failed with an error." (ERROR) → Email: subject "Set HTOC Scanner Identifier Failed", body = TC error message, to \`jaytlin.askew@hhs.gov\` → Loop continues (does not halt the remaining indicators)

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7. Step Details

Step 1 — ThreatConnect Get Indicators

App: TCPB - Threatconnect Get Indicators v1 (v1.0.9)

Action: TQL

Output: `tc.indicators.entity` (TCEntityArray) and `tc.indicators.summary` (StringArray)

Fail on no results: Yes — triggers the error path if no new indicators are found

Also connects to: Indicator List Summary Logger (logs the full summary list for each run)

Step 2 — Iterator

App: Iterator (v1.0.0)

Input: `tc.indicators.entity` array from Step 1

Output: `currentIndicator` (TCEntity) — one indicator per loop iteration

Loop behavior: Iterates until all entities are processed; on loop completion (Pass) advances to the Success step; on failure sends the "playbook failed" email

Step 3 — ThreatConnect Tags (Get)

App: TCPB - Threatconnect Tags TI v2 (v2.0.7)

Action: Get

Entity: `currentIndicator` from the Iterator

Fail on no results: No — indicators without existing tags proceed normally

Step 4 — ThreatConnect Set HTOC Scanner Identifier

App: TCPB - Threatconnect Tags TI v2 (v2.0.7)

Action: Add

Tag applied: `CDN PB`

Apply to all: Yes

Entity: `currentIndicator`

Fail on error: Yes — failed tag additions route to the tag-failure email path

Purpose: Marks the indicator as processed so the TQL query excludes it in future runs

Step 5 — JMESPath

App: TCPB - JMESPath v3.0 (v3.0.12)

Input: `tc.response.json.raw` from Step 4 (tag response JSON)

Expressions:

`[0].summary` → output variable `indicator` (String)

`[0]` → output variable `indicatorData` (String)

Fail on expression error: Yes

Step 6 — DataStore (POST)

App: TCPB - DataStore v4 (v4.0.5)

Action: POST

Domain: Local

Data type: JSON

Key (rid): `indicator` (the indicator summary from JMESPath)

Data: `{"indicatorData": <indicatorData from JMESPath>}`

Purpose: Persists indicator metadata to the ThreatConnect Local DataStore for downstream reference

Step 7 — Complete Success Logger

App: Logger (v1.0.0)

Level: INFO

Message: "Process completed with no issues"

Step 8 — Logger (Loop End)

App: Logger (v1.0.0)

Level: INFO

Message: "Tagging process completed."

Behavior: Triggers the EndLoop connection back to the Iterator

Step 1a — Indicator List Summary Logger

App: Logger (v1.0.0)

Level: INFO

Message: Contents of `tc.indicators.summary` (StringArray) — logs all indicator summaries returned by the query

Purpose: Provides a per-run audit log of which indicators were discovered before the loop begins

Step 9 — Success Logger

App: Logger (v1.0.0)

Level: INFO

Message: "Get CDN Scanning Indicators by ASN Completed Successfully!"

Output: `logger.content` — returned as the WebHook HTTP response body when triggered via webhook

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8. Email Alerts

All failure notifications are sent to **`jaytlin.askew@hhs.gov`**. Update this address in all three Send Email steps if the playbook owner changes.

Alert	Subject	Body	Trigger
Get Indicators failed	`GET Indicator Process Failed`	Logger error content	Get Indicators step fails

Alert	Subject	Body	Trigger
Iterator / no results	`Get CDN Scanning Indicators by ASN Failed.`	Static: "Get CDN Scanning Indicators by ASN Failed."	Iterator step fails
Tag add failed	`Set HTOC Scanner Identifier Failed`	ThreatConnect error message from tag step	Tag Add step fails

Connection type: SMTPS (Implicit TLS)

Timeout: 30 seconds

Send one: Yes (single email per event, not one per indicator)

Fail on error: No (email failures do not halt the playbook)

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9. Modifying the Playbook

Adding or Removing Monitored ASNs

Open the playbook in the ThreatConnect editor.

Click on the **ThreatConnect Get Indicators** step.

Update the `addressASN IN (...)` clause in the TQL field.

Save and re-activate the playbook.

Changing the Max Results per Run

The `max_results` parameter in **ThreatConnect Get Indicators** is currently set to **5**. Increase this value to process more indicators per execution. Be aware that higher values increase execution time and the risk of hitting the 5-minute WebHook timeout.

Changing the Scanner Tag Name

The tag `CDN PB` is the marker used to prevent reprocessing. To rename it:

Update the `name` parameter in **ThreatConnect Set HTOC Scanner Identifier**.

Update the `NOT hasTag(summary IN ("CDN PB"))` clause in the TQL.

Bulk-update existing tagged indicators in ThreatConnect before reactivating to avoid reprocessing.

> **Warning:** If the tag name in the TQL exclusion filter is not updated to match the new tag being applied, indicators will be reprocessed on every execution.

Updating the Notification Email

Update the `recipient` field in all three **Send Email** steps to the new owner's address.

Managing Triggers

Every 12 Hours (6AM-6PM): This is the production trigger. It should always be active during normal operations.

WebHook Trigger: For testing and manual one-off runs only. Activate when needed and confirm the 12-hour timer remains the primary trigger.

Every 1 Minute (Testing Trigger): For development/debugging only. Disable immediately after use.

To toggle a trigger, open the playbook editor, click the trigger, and toggle the Active switch.

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10. Troubleshooting

Symptom	Likely Cause	Resolution
Email: "GET Indicator Process Failed"	TQL syntax error, Mandiant feed unavailable, or no matching indicators with fail_on_no_results=true	Review the TQL in Get Indicators; verify the Mandiant feed is active and syncing
Email: "Get CDN Scanning Indicators by ASN Failed."	Iterator received an empty or failed entity array	Confirms no new untagged CDN indicators were found for the monitored ASNs — may be normal on low-activity days
Email: "Set HTOC Scanner Identifier Failed"	Tag add API error — possible permissions issue or indicator locked	Check ThreatConnect permissions for the playbook service account; review the TC error message in the email body
Indicators being reprocessed repeatedly	`CDN PB` tag is not persisting on indicators	Verify the tag add step is not failing silently; confirm the tag name in the TQL matches exactly (case-sensitive)
Playbook not triggering	WebHook trigger inactive or webhook URL changed	Verify WebHook Trigger is active in the playbook editor; confirm the calling system has the correct webhook URL

Symptom	Likely Cause	Resolution
DataStore POST errors	Local DataStore unavailable or data format issue	Check ThreatConnect DataStore status; verify JMESPath is producing valid `indicatorData` JSON
Playbook times out	Processing via WebHook exceeds 5-minute timeout	The 12-hour timer trigger has no HTTP timeout constraint and is the preferred production trigger; use it instead of the webhook for normal operations

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11. Maintenance Notes

Tag name (`CDN PB`): This is the production tag name used to mark processed indicators. Do not rename without simultaneously updating the TQL exclusion filter.

Max results = 1: The playbook processes one new indicator per execution. With the 12-hour schedule this means up to 2 new indicators tagged per day. Increase `max_results` if faster catch-up is needed, keeping the 5-minute WebHook timeout in mind if using the webhook trigger.

Active trigger is Every 12 Hours: The production schedule runs at 6:00 AM and 6:00 PM. The WebHook trigger is for testing only and should not be relied upon as the primary execution method.

DataStore use: The Local DataStore is being used to persist indicator data. Ensure periodic review or cleanup of DataStore records to prevent unbounded growth.

Version history: This is version 1.8. Export a new `.pbxz` backup from ThreatConnect after any significant changes.

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12. Related Documents

ThreatConnect Playbook documentation: [ThreatConnect Playbook Apps](https://docs.threatconnect.com)

Mandiant Advantage Threat Intelligence feed documentation